
ReTrace: Model-Response Watermarking for Diffusion Language Models

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Abstract

1 We study a watermark for masked diffusion language models whose detector
2 reruns the model instead of inspecting token identities alone. ReTrace compares
3 each observed token’s likelihood under two replayable context ablations; a keyed
4 orientation determines which response sign counts as a match. Reproducible,
5 orientation-independent carrier selection gives the match count an exact conditional
6 $\text{Binomial}(n, \frac{1}{2})$ null. During blockwise decoding, bounded rejection sampling
7 favors model proposals with the requested sign and falls back to an unchecked
8 draw. On LLaDA-8B-Instruct, the best evaluated 512-token settings obtain 0.70–
9 0.80 TPR at 1% FPR with matched-control GPT-2 perplexity ratios of $0.87\times$ –
10 $0.74\times$. The channel is useful but costly: detection requires 144 masked sequence
11 evaluations, trails strong token-based baselines, and drops from 0.35 to 0.05 after
12 re-denoising 10% of positions. ReTrace establishes a model-response verification
13 channel while exposing its quality, compute, and robustness limits.

14 1 Introduction

15 Language-model watermarks usually encode a keyed pattern in token identities: red–green methods
16 bias keyed vocabulary subsets [11], sampling-based methods couple choices to pseudorandom-
17 ness [13], and recent diffusion language model (DLM) methods exploit unresolved context or
18 denoising order [9]. Their detectors inspect the text and key without rerunning the generator. We
19 instead ask whether the model itself can be the verification instrument.

20 A masked DLM can revisit a completed text: hide selected context and observe how the likelihood of
21 an emitted token changes. ReTrace compares two replayable context ablations, giving each candidate
22 token a signed response. A secret orientation specifies which sign counts as a match. Under a
23 key-independent text, orientations are independent fair bits; the match count therefore has an exact
24 conditional $\text{Binomial}(n, \frac{1}{2})$ null, without a calibration corpus or Gaussian approximation.

25 The embedding opportunity is *local conditional ambiguity*. Several plausible tokens may coexist at
26 a masked position yet react with opposite signs to the two ablations (Figure 1). If both signs carry
27 probability mass, bounded resampling can prefer the requested sign while accepting only model
28 proposals. Small per-position biases then accumulate in the detector. A sharp or one-sided conditional
29 offers no choice and is skipped rather than forced.

30 ReTrace comprises four modules (Figure 2): the RRB, BCG, RAR, and RRV. The RAR uses an
31 unconditional fallback after R failed guided proposals and a key-free plausibility floor that excludes
32 proposals far below the live argmax. This is rejection sampling: it changes the output distribution,
33 although retries reuse one probability row and every committed token is drawn from the model.

34 ReTrace is not intended to dominate earlier watermarks on every axis. Its advantage is a different
35 operating constraint. Applying a prefix-based watermark to a DLM can require strict left-to-right

Where does ReTrace find room for a watermark?

illustrative probability row

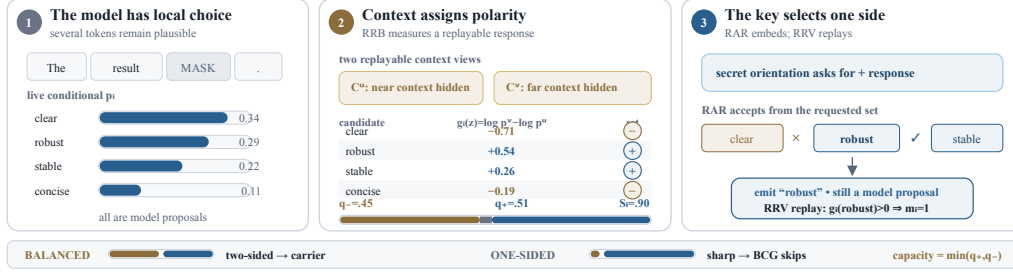


Figure 1: **Where the watermark channel comes from.** A non-degenerate DLM conditional contains several plausible proposals. Two replayable context views partition that support by response sign into masses q_+ and q_- . The key requests one side; RAR can select from it and RRV can later remeasure the emitted token’s sign. Values are illustrative. Capacity is limited by the weaker sign mass, so BCG rejects sharp or one-sided positions.

36 decoding, while order-guided methods deliberately change which mask is resolved next. ReTrace
 37 preserves the model’s native block and confidence schedule and changes only the sampled proposal
 38 at admitted positions. More importantly, its detector tests a keyed relation between the final text
 39 and a particular model’s contextual response, rather than a key–token hash alone. The result is
 40 checkpoint-bound and expensive, but it provides exact conditional false-positive control even though
 41 carriers are selected from the observed text and model predictions.

42 Our contributions are: (i) model-response verification with reproducible carriers and an exact finite-
 43 sample null; (ii) a response-guided embedder with an explicit detection–quality–compute trade-
 44 off; and (iii) an evaluation on LLaDA-8B-Instruct that reports matched-control perplexity, model-
 45 evaluation cost, detector checks, and post-editing failure modes. At 512 tokens, the strongest evaluated
 46 settings reach 0.70–0.80 TPR at 1% FPR with measured PPL ratios of $0.87\times$ – $0.74\times$, but require
 47 144 masked evaluations for verification and are fragile to editing. ReTrace is therefore evidence that
 48 a model-response channel exists, not a state-of-the-art detector.

49 2 Related Work

50 **Diffusion language models.** Discrete and masked diffusion objectives extend diffusion to text
 51 [3, 14, 16, 18, 19] and now support instruction-following systems such as LLaDA and Dream
 52 [15, 23, 25]. Practical decoders complete blocks from left to right and use confidence to select
 53 positions within a block [2]; token order affects quality and efficiency [10, 20, 21]. ReTrace preserves
 54 this reference schedule and uses masked-token reconstruction only to define a replayable response.

55 **Text watermarking.** Autoregressive watermarks bias keyed token sets [11] or couple sampling to
 56 keyed randomness [1, 7, 13]; subsequent work studies reliability, robustness, and undetectability [6,
 57 12, 24]. These token-only detectors are inexpensive but do not transfer unchanged to unresolved
 58 DLM contexts. DLM-specific methods instead remove prefix dependence [5], average a red–green
 59 bias over unresolved context [8], use keyed sampling [4], or exploit decoding order [9, 17, 22]. We
 60 reproduce dgMARK locally because it offers a strong order-steering baseline with zero-inference
 61 detection. ReTrace differs at verification: the detector must rerun the exact DLM checkpoint under
 62 keyed corruptions. This supplies a model-response statistic but incurs substantial cost.

63 The distinction is operational rather than cosmetic. A left-to-right green-list method reintroduces a
 64 prefix solely to make its key–token partition well defined; dgMARK instead uses the DLM-specific
 65 freedom to choose which position is committed next. ReTrace leaves both the block partition
 66 and confidence-based position choice untouched. Its carrier is a counterfactual response of the
 67 model to visible context, so the selected positions may be text- and model-dependent while the
 68 secret orientation remains independent. This separation is what permits an exact conditional test
 69 after adaptive, model-driven carrier selection. It also explains why ReTrace should be viewed as
 70 complementary to inexpensive token hashes: the added semantic binding to a checkpoint is useful
 71 only when model-assisted verification and exact replay are acceptable deployment costs.

3 Background and Problem Setup

Let x be a prompt and $y = (y_1, \dots, y_N)$ a continuation partitioned into blocks. We study LLaDA’s semi-autoregressive schedule: blocks finish left to right, while diffusion resolves masks within the active block [2, 15]. One sequence evaluation returns a categorical distribution at every unresolved position, and the reference confidence rule decides which positions to commit.

The embedder receives the model, prompt, secret key K , and target bits. The verifier later receives only the completed sequence, the same model, K , and public parameters; it does not receive logits, random seeds, or commit order. A nonce derives a document key $K_d = \text{HMAC}(K, \text{nonce}_d)$. The null covers any text generated independently of K_d .

Threat model and scope. The model and algorithm may be public; secrecy rests in K . We require false-positive control for arbitrary fixed text independent of the document key, including machine text from another sampler. We do not claim security after key disclosure, against an adaptive adversary with detector-query access, or when verification uses a numerically different checkpoint. The latter restriction is deliberate: ReTrace binds its evidence to the contextual behavior of a specified model, but sacrifices the portability of token-only detectors.

Design requirements. Three constraints shape the construction. *Replayability* requires every scored context to be recoverable from the final text. *Orientation independence* requires carrier selection to ignore which response sign the key will reward; otherwise the null can be biased. *Schedule fidelity* requires embedding to retain the reference block and confidence order. We evaluate power at fixed FPR, paired quality against the same base sampler, and generation and verification model evaluations.

4 ReTrace

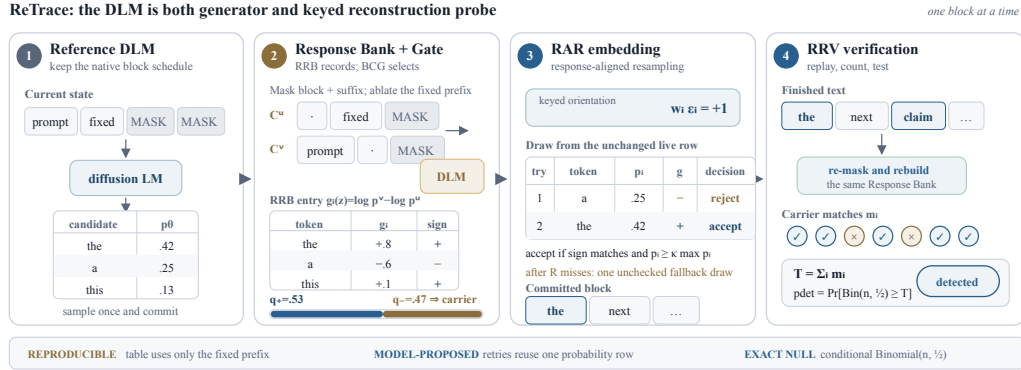


Figure 2: Overall ReTrace pipeline. The reference DLM retains its block and confidence schedule. RRB records context-ablation responses; BCG admits positions with mass on both signs; and RAR retries from the unchanged live conditional. For detection, RRV re-masks each completed block, reconstructs the same bank and carriers, counts keyed sign matches T , and returns an exact binomial p -value. Verification needs the final text, model, public parameters, and key, but no generation trajectory or rejected samples.

ReTrace must reproduce at verification time every decision that determines which tokens are scored. This requirement divides the method into four modules. The *Reproducible Response Bank* (RRB) records how the DLM responds to a fixed family of context ablations (§4.1). The *Balanced Carrier Gate* (BCG) admits only positions with usable probability mass on both response signs. The *Response-Aligned Resampler* (RAR) embeds evidence without changing the reference block schedule (§4.2), and the *Response-Replay Verifier* (RRV) reconstructs the RRB from the completed text and applies an exact test (§4.3).

4.1 Reproducible Response Bank and Balanced Carrier Gate

For block B_b , let F_b be the prompt and completed prefix. ReTrace constructs L equal-size near, far, and keyed pseudorandom ablations $D_{b,\ell} \subset F_b$. It masks $D_{b,\ell}$, the current block, and the suffix; one model evaluation then yields, for all $i \in B_b$ and vocabulary items v ,

$$\ell_{i,\ell}(v) = \log p_\theta(v \mid F_b \setminus D_{b,\ell}; B_b \text{ and suffix masked}). \quad (1)$$

Re-masking the block and suffix removes tokens unavailable when its bank was built. The verifier can therefore restore the same state from the final text; no trajectory is needed. Masking only the ablated prefix would fail: tokens committed later in the same block would leak into verification-time logits, while the future suffix was entirely invisible during embedding. Block-and-suffix masking is therefore the mechanism that turns a transient generation-time response into a document-replayable statistic.

For an unordered pair of patterns (u, v) , define

$$g_i^{u,v}(z) = \ell_{i,v}(z) - \ell_{i,u}(z). \quad (2)$$

Let p_i^{base} denote the model distribution when B_b and the suffix are masked but no context pattern is ablated. The probability mass on each sign is

$$q_{i,+}^{u,v} = \sum_z p_i^{\text{base}}(z) \mathbf{1}[g_i^{u,v}(z) > 0], \quad q_{i,-}^{u,v} = \sum_z p_i^{\text{base}}(z) \mathbf{1}[g_i^{u,v}(z) < 0], \quad (3)$$

and the two-sided score is

$$S_i^{u,v} = 2 \min(q_{i,+}^{u,v}, q_{i,-}^{u,v}). \quad (4)$$

ReTrace chooses, independently at each position, the pair with the largest $S_i^{u,v}$ and stores its contrast as g_i in the RRB. The *Balanced Carrier Gate* then admits position i when $S_i > s_{\min}$; an optional per-block cap retains the largest scores. Because $S_i^{u,v}$ is unchanged when u and v are swapped, the BCG neither observes nor favors the requested orientation. Both pair selection and gating are recomputable from the block-masked state.

After the BCG admits a position, the key supplies an orientation $\epsilon_i \in \{-1, +1\}$ from a PRF domain separate from pattern generation, pair selection, carrier assignment, and payload grouping. A target $w_i \in \{-1, +1\}$ represents the desired provenance or message bit. For a candidate z , the shared embedding-and-verification predicate is

$$m_i(z) = \mathbf{1}[w_i \epsilon_i g_i(z) > 0]. \quad (5)$$

If $g_i(z) = 0$ exactly, a second domain-separated fair bit defines $m_i(z)$. Such ties cannot be steered by the RAR, but the tie rule is required for the RRV’s exact null.

Because S_i uses the reproducible block-masked distribution while generation uses an evolving live conditional, it is a symmetric proxy rather than a guarantee of easy steering. Best-pair selection raises the measured mean score from 0.28 for a fixed near/far pair to 0.45.

4.2 Response-Aligned Resampling

Algorithm 1 ReTrace embedding and detection. RAR and RRV share BUILD-RRB; guided retries reuse the current probability row.

Input: prompt $x_{<B_1}$, key K , retry budget R , floor κ , threshold $s_{\min}$ **Output:** generated text y or p_{det}

BUILD-RRB(x, b, K) — Reproducible Response Bank

- 1 Mask the current block B_b and suffix; construct L keyed ablations of the fixed prefix.
- 2 Evaluate the L ablated sequences and one unablated sequence with the DLM.
- 3 For every $i \in B_b$, select $(u, v) = \arg \max_{u < v} S_i^{u,v}$ and store $g_i = \ell_{i,v} - \ell_{i,u}$ and $S_i = 2 \min(q_{i,+}, q_{i,-})$.

EMBED(x, K, R, κ) — Response-Aligned Resampler

- 4 **for** each block B_b in reference order **do**; compute $\mathcal{R}_b \leftarrow \text{BUILD-RRB}(x, b, K)$.
- 5 **while** B_b contains masks **do**; let the reference confidence rule select positions and obtain each live row p_i .
- 6 **if** $S_i \leq s_{\min}$, draw $z \sim p_i$ once.
- 7 **else** draw at most R proposals from p_i ; accept the first z satisfying $w_i \epsilon_i g_i(z) > 0$ and $p_i(z) \geq \kappa \max_v p_i(v)$.
- 8 **if** no guided proposal is accepted, draw one fresh unchecked $z \sim p_i$; commit z .

VERIFY(y, K) — Response-Replay Verifier

- 9 Re-mask each completed block and suffix; rebuild $\mathcal{R}_b$ with BUILD-RRB(y, b, K).
 - 10 Apply the BCG, compute m_i at admitted carriers, and return the exact binomial tail in Eq. (10).
-

130 Within each block, RAR retains the reference confidence order and live conditional p_i ; positions
 131 rejected by BCG are sampled normally. At an admitted carrier define

$$A_i = \left\{ z : w_i \epsilon_i g_i(z) > 0 \text{ and } p_i(z) \geq \kappa \max_v p_i(v) \right\}. \quad (6)$$

132 RAR draws from p_i up to R times and commits the first member of A_i ; after R misses it commits
 133 one fresh unchecked draw. The row is unchanged, so retries require no model evaluation.

134 This procedure changes the sampling distribution. Let $a_i = p_i(A_i)$ be the acceptance mass and let

$$q_i = \sum_z p_i(z) m_i(z) \quad (7)$$

135 where $m_i(z)$ applies Eq. (5) and the tie rule to candidate z . Thus q_i is the probability that an
 136 unconditional draw is scored as a detector hit, irrespective of the plausibility floor. The committed-
 137 token distribution is

$$(1 - (1 - a_i)^R) p_i(\cdot | A_i) + (1 - a_i)^R p_i, \quad (8)$$

138 and its detector-hit probability is

$$\Pr[m_i = 1] = 1 - (1 - a_i)^R (1 - q_i). \quad (9)$$

139 When $\kappa = 0$ and the conditional assigns no mass to exact ties, $a_i = q_i$ and Eq. (9) reduces to
 140 $1 - (1 - q_i)^{R+1}$. The distinction matters with a floor or tie rule because an unchecked fallback may
 141 score as a hit without belonging to A_i .

142 The floor bounds each guided proposal’s model NLL to within $-\log \kappa$ of the live argmax; it neither
 143 constrains the fallback nor guarantees semantic quality. Being model-proposed is a support statement,
 144 not a distortion-free claim.

145 Together, the BCG, unconditional fallback, and plausibility floor determine the RAR’s empirical
 146 detection–quality trade-off.

147 4.3 Response-Replay Verification and exact null

148 Given completed text y , RRV re-masks each block and suffix, rebuilds its RRB from the fixed
 149 prefix, reapplies BCG, and evaluates $m_i = m_i(y_i)$ at admitted positions. Rejected proposals and the
 150 generation trajectory are irrelevant.

151 For provenance-only detection, all admitted carriers form a presence pool P with $w_i = +1$. If
 152 a payload is enabled, the implementation can reserve a synchronization subset for presence and
 153 partition the remaining carriers by payload bit. These groups must remain separate: aggregating a
 154 balanced message would cancel positive and negative targets. The presence statistic and its one-sided
 155 p -value are

$$T = \sum_{i \in P} m_i, \quad p_{\text{det}} = \Pr \left[\text{Binomial}(|P|, \frac{1}{2}) \geq T \right]. \quad (10)$$

156 **Proposition 1** (Conditional exact null). *Assume the PRF domains above behave as independent*
 157 *random functions. Fix any text y generated independently of the document key, and condition on*
 158 *the carrier set, selected pairs, contrast values, payload roles, and target bits. The orientation bits*
 159 *(and tie bits where needed) remain independent fair coins. Consequently, the m_i are independent*
 160 *Bernoulli($\frac{1}{2}$) variables and $T \sim \text{Binomial}(|P|, \frac{1}{2})$ exactly.*

161 The result requires neither a text distribution nor an asymptotic approximation, but it does require
 162 key-independent null text and correct domain separation, implemented with distinct HMAC labels.
 163 The conditioning in Proposition 1 is important. The carrier count, selected ablation pair, and contrast
 164 magnitude may all depend arbitrarily on the text and model. Exactness survives because these
 165 quantities are fixed before the independent orientation bit is consulted. A gate based on the requested
 166 sign would destroy this symmetry and make the nominal binomial tail anti-conservative.

167 5 Experiments

168 Following dgMARK’s evaluation organization, we first specify data, generation, metrics, and base-
 169 lines, then analyze detectability, quality, distributional effects, and robustness.

Method	Setting	PPL ratio ↓	TPR at FPR ↑		Gen. eval.	Det. eval.
			1%	0.1%		
KGW [11]	$\delta = 1, 256$ tok	$1.03\times$	0.93	0.73	256	0
dgMARK [9]	$k = 1, 256$ tok	$1.55\times$	0.74	0.62	256	0
dgMARK [9]	+3-beam, 256 tok	$1.23\times$	0.86	0.82	~ 1024	0
ReTrace	$R = 1, 512$ tok	$0.97\times$	0.43	0.20	656	144
ReTrace	$R = 8, \kappa = 0.1, 512$ tok	$0.87\times$	0.70	0.60	656	144
ReTrace	$R = 16, \kappa = 0.05, 512$ tok	$0.74\times$	0.80	0.60	656	144

Table 1: **Detection, matched-control perplexity, and compute.** All rows are local reproductions on one checkpoint and prompt protocol, but ReTrace uses 512-token outputs while token-based baselines use 256. PPL ratio compares with the matched unwatermarked sampler; below one is not evidence of better semantics. Eval. counts sequence evaluations. Principal rows use $n = 30$ documents ($n = 20$ for $R = 16$); dgMARK uses $n = 50$.

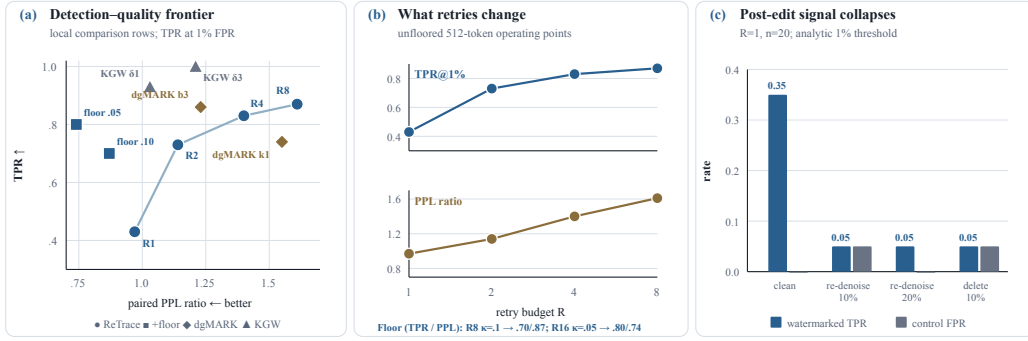


Figure 3: **Detection, quality, and the principal failure mode.** (a) Local detection-PPL operating points; differing output lengths make this a system comparison, not a controlled ranking. (b) Without a floor, retries improve TPR but raise PPL; floor settings recover lower-PPL operating points. (c) On 20 $R = 1$ documents, re-denoising or deleting 10% of positions reduces TPR from 0.35 to 0.05.

5.1 Experimental setup

Dataset, model, and environment. We use GSAI-ML/LLaDA-8B-Instruct in fp16 on one TITAN RTX. C4 realnewslike prompts are truncated to 300 characters, matching dgMARK’s protocol. Unless stated otherwise, generation produces 512 tokens in 32-token blocks with one denoising step per token. Principal ReTrace results use 20–40 prompts; the comparison table reports every sample size rather than treating these modest runs as population-level estimates.

Sampling and ReTrace configuration. The reference sampler uses full-vocabulary multinomial sampling at temperature 0.8 and the native confidence schedule. ReTrace adds eight ablation patterns, threshold $s_{\min} = 0.5$, retry budget R , and optional plausibility floor κ . Every document uses an HMAC-derived key. We test provenance detection ($w_i = +1$), not payload recovery.

Metrics and baselines. Detectability is TPR at analytic FPR thresholds using Eq. (10). Quality is GPT-2-large perplexity paired with a control using the same schedule and temperature; we report $\exp(\text{mean}(\text{NLL}_{\text{wm}} - \text{NLL}_{\text{ctrl}}))$. Pairs failing the decoded-length rule are excluded. Compute is the number of full sequence evaluations, independent of batching. We locally reproduce KGW [11] under strictly left-to-right DLM decoding and dgMARK [9] under confidence decoding.

5.2 Experimental analyses

Watermark detectability. At 512 tokens, $R = 1$ reaches 0.43 TPR at 1% FPR. On 30 held-out prompts, increasing R from 2 to 4 to 8 raises TPR from 0.73 to 0.83 to 0.87 (Figure 3b). This agrees with Eq. (9): repeated draws increase the chance of reaching the requested response set.

Effect of sampling strength. Retries require no extra DLM evaluations because they reuse the current row, but unfloored PPL ratios rise from $1.14\times$ at $R = 2$ to $1.40\times$ at $R = 4$ and $1.61\times$ at $R = 8$. The plausibility floor trades some signal for quality: $(R, \kappa) = (8, 0.1)$ obtains 0.70 TPR with a $0.87\times$ ratio, while $(16, 0.05)$ obtains 0.80 TPR with a $0.74\times$ ratio. These below-one ratios mean only that GPT-2-large prefers the selected high-probability tokens; they do not establish better semantics.

Effect of generation length. Longer outputs supply more carrier trials. At $R = 1$, moving from 256 to 512 tokens raises TPR at 1% FPR from 0.25 to 0.40 in a separate run. At 1024 tokens, $(R, \kappa) = (8, 0.1)$ reaches 0.75 TPR at 1% FPR and 0.69 at 0.1%, with a $0.83\times$ PPL ratio. This setting is less conclusive: only 8 of 16 pairs pass the decoded-length rule, and bigram repetition rises from 0.565 in the control to 0.623. Evidence therefore scales with length, but generation stability becomes a confound.

Direct effect on the output distribution. Unlike dgMARK’s order preference, RAR directly performs rejection sampling. Equation (8) states the changed distribution, and the floor constrains only guided acceptances. The unchecked fallback preserves termination but can emit any live proposal. Thus *model-proposed* does not mean distribution preserving.

Text generation quality. For $(R, \kappa) = (8, 0.1)$, decoded bigram repetition is 0.33 versus 0.36 for its matched control. This and PPL provide only surface-level evidence; no downstream-task or human evaluation was run. We therefore report a measured PPL operating point, not an imperceptibility claim.

Comparison and computational overhead. KGW reaches 0.93 TPR at 1% FPR and dgMARK reaches 0.74 (0.86 with three-beam lookahead), both above comparable ReTrace settings (Table 1). ReTrace generation costs $656/512 = 1.28\times$ the reference sequence-evaluation budget and verification costs 144 masked evaluations; KGW and dgMARK detection uses none. KGW, however, requires left-to-right generation, and its repeated bigram events must be deduplicated; without that correction its control measured 37% FPR. The dgMARK three-beam row costs roughly four generation evaluations per token. These are local reproductions, not performance ceilings.

When is ReTrace preferable? Table 1 shows no universal win: token-based detectors are cheaper and usually stronger. ReTrace instead improves two constrained operating points. First, it keeps native blockwise decoding. The KGW-compatible left-to-right control itself incurs a $1.17\times$ PPL cost relative to block decoding and exhibits high repetition, while dgMARK obtains its signal by changing the position order. Second, ReTrace’s model-driven carrier selection remains compatible with an exact finite-sample null; no empirical threshold calibration is needed. Relative to dgMARK three-beam, ReTrace also uses a lower generation budget ($1.28\times$ versus roughly $4\times$ reference decoding), although its 1%-FPR TPR is lower (0.80 versus 0.86) and its verifier is far more expensive. ReTrace is thus preferable only when schedule fidelity and model-bound exact verification outweigh detector cost and edit robustness.

Detector checks. The exact-null result is analytical. In finite control arms, the 30-document retry run and 40-document $R = 1$ run each yield zero detections at the 1% threshold, too few samples to estimate a 1% FPR precisely. On the same cached $R = 1$ outputs, gated sign count and two likelihood-ratio variants each reach 0.38 TPR; counting all positions falls to 0.33. We retain the simpler gated binomial detector.

Robustness to post-editing. On 20 $R = 1$ documents, same-model argmax re-denoising of a random 10% of positions reduces TPR at 1% FPR from 0.35 to 0.05; deleting 10% has the same measured effect. Early edits alter the fixed prefix used by later RRBs. ReTrace therefore does not currently support claims of edit robustness.

Why not post-hoc substitution? Across 36 settings that replace tokens after ordinary generation, the median selected position has one admissible token at likelihood budget $\tau = 3$; the mean is 1.6 at $\tau = 3$ and 5.6 at $\tau = 6$. The best setting within a $1.35\times$ PPL budget reaches only 0.10 TPR at 1% FPR. This evaluated model’s sharp conditionals leave little room for forced replacement, motivating steering during sampling and an unchecked fallback.

240 6 Discussion

Method	Watermark evidence	DLM schedule	Gen. budget	Verifier	Principal cost
KGW [11]	keyed token subset	left-to-right adaptation	1.0×	text only	loses block decoding
dgMARK [9]	keyed parity via order	order is steered	1.0–4.0×	text only	changes commit order
ReTrace	contextual model response	preserved	1.28×	144 model eval.	edit-fragile, checkpoint-bound

Table 2: **The methods optimize different constraints.** Budgets are local sequence-evaluation ratios at the settings in Table 1; the dgMARK range spans one-candidate and three-beam decoding. Preserved means that ReTrace leaves the reference block partition and confidence-based position order unchanged.

241 **Schedule fidelity.** The clearest advantage over prefix-based adaptation is that ReTrace does not
 242 serialize the DLM. In our control, the left-to-right schedule needed by KGW raises PPL by $1.17\times$
 243 relative to block decoding and produces high repetition before any watermark strength is considered.
 244 ReTrace also differs from dgMARK: dgMARK turns order sensitivity into its channel, whereas
 245 ReTrace treats the reference order as an invariant and acts only after a live probability row exists. This
 246 distinction matters when the production decoder has a validated latency–quality schedule that should
 247 not be changed. It is not a free advantage: constructing the RRB adds 28% generation evaluations.

248 **Model-bound evidence.** A token-only detector asks whether token identities correlate with a key.
 249 RRV asks a different question: whether the emitted tokens exhibit keyed responses when a specified
 250 model reconstructs them under counterfactual context views. The evidence is therefore bound jointly
 251 to text, context, and checkpoint. This can be useful for provenance settings where the verifier controls
 252 the model artifact and wants evidence tied to its behavior, rather than to any generator capable of
 253 reproducing a token histogram. The same property is a liability for open verification because a
 254 different checkpoint need not reproduce the responses.

255 **Exactness after adaptive carrier discovery.** BCG chooses carriers and ablation pairs from the
 256 completed prefix and model probabilities; the number and identity of trials are consequently data
 257 dependent. Nevertheless, after conditioning on those choices, the orientation bits remain independent
 258 fair coins and the binomial tail is exact. The advantage is not merely replacing a z -score with a
 259 binomial function: it is preserving finite-sample validity after model-driven carrier discovery. The
 260 measured KGW failure without bigram deduplication illustrates why dependence and trial definition
 261 cannot be treated as implementation details.

262 **Boundary of the claim.** Table 2 makes clear that ReTrace is not uniformly superior. KGW and
 263 dgMARK have zero-inference detectors, stronger local TPR, and token-local evidence that is easier
 264 to synchronize after edits. ReTrace is preferable only when preserving the native DLM schedule,
 265 obtaining a replayable model-response statistic, and retaining exact conditional calibration are worth
 266 an expensive checkpoint-specific verifier. A practical system could also combine these channels: a
 267 cheap token-local watermark for routine screening and RRV as a higher-cost confirmation layer. This
 268 composition is a design direction, not an evaluated result.

269 **Limitations.** Results cover one checkpoint, one corpus, small samples, and automatic quality
 270 proxies. Cross-model generality, human or downstream quality, scalable verification, adaptive attacks,
 271 and content-anchored synchronization remain open.

272 7 Conclusion

273 ReTrace shows that keyed, replayable changes in a DLM’s predictions can form a watermark channel
 274 with an exact conditional null. Balanced gating identifies usable positions, resampling embeds sign
 275 evidence, and re-masking reconstructs it without a generation log. The current operating point is
 276 a proof of concept: verification is expensive, evidence is edit-fragile, and evaluation covers one
 277 checkpoint. Scaling response replay and anchoring challenges to content are the principal next steps.

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